# iregi-rust benchmark

Batched one-loop tensor reduction in Rust versus IREGI and MadLoop

2026-10-23 · AMD EPYC 9754 · CPU 0 · identified kernel snapshots

**Reading guide.** Every timing ratio is *iregi-rust time ÷ reference time*. Green means faster, white parity and red slower; timings are per phase-space point. Gray cells are unmeasured; amber marks pending or inconclusive evidence. Source inspection does not establish numerical agreement. The newer 5c9e3dd interpreter snapshot passes all eight numerical checks: six regressions and two inconclusive noisy timing rows. Historical 4c32ece runs remain separate. Full performance acceptance is unmet; subsequent default-stack/generalized-PV/vector changes remain unmeasured.

**Acceptance gates**

| | | |
|---|---|---|
| **Correctness** | Agreement with original IREGI and isolated MadLoop graphs | **Pending** |
| **Kernel performance** | Equal or better on identical inputs; three separate runs | **Unmet** |
| **Complete evaluation** | Report preparation, numerator, reduction and masters separately | **Unmeasured** |

## 1 Historical kernel baseline: 4c32ece

Table 1: **Massive bubble/triangle, binary64, hybrid mode.** Rust and IREGI total medians in µs/point. Ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

| Case | Rank | Interpreter Batch 1 | Interpreter Batch 32 | Native Batch 1 | Native Batch 32 |
|---|---|---|---|---|---|
| Bubble | 0 | 3.176 µs<br>IREGI 1.988 µs · ×1.598<br>99% CI 1.585–1.607 | 1.708 µs<br>IREGI 1.885 µs · ×0.906<br>99% CI 0.902–0.911 | 0.936 µs<br>IREGI 0.943 µs · ×0.993<br>99% CI 0.987–1.003 | 0.903 µs<br>IREGI 0.878 µs · ×1.031<br>99% CI 1.024–1.042 |
| Bubble | 1 | 9.552 µs<br>IREGI 5.707 µs · ×1.670<br>99% CI 1.655–1.688 | 5.831 µs<br>IREGI 5.748 µs · ×1.015<br>99% CI 1.008–1.019 | 4.663 µs<br>IREGI 3.044 µs · ×1.534<br>99% CI 1.526–1.549 | 3.231 µs<br>IREGI 3.029 µs · ×1.069<br>99% CI 1.062–1.081 |
| Triangle | 0 | 29.152 µs<br>IREGI 8.496 µs · ×3.446<br>99% CI 3.415–3.454 | 18.229 µs<br>IREGI 8.433 µs · ×2.160<br>99% CI 2.147–2.172 | 4.227 µs<br>IREGI 4.968 µs · ×0.849<br>99% CI 0.844–0.858 | 8.405 µs<br>IREGI 4.899 µs · ×1.715<br>99% CI 1.704–1.723 |
| Triangle | 1 | 38.505 µs<br>IREGI 18.297 µs · ×2.101<br>99% CI 2.088–2.108 | 25.634 µs<br>IREGI 18.224 µs · ×1.408<br>99% CI 1.400–1.413 | 8.809 µs<br>IREGI 10.418 µs · ×0.850<br>99% CI 0.844–0.854 | 12.461 µs<br>IREGI 10.213 µs · ×1.220<br>99% CI 1.206–1.228 |



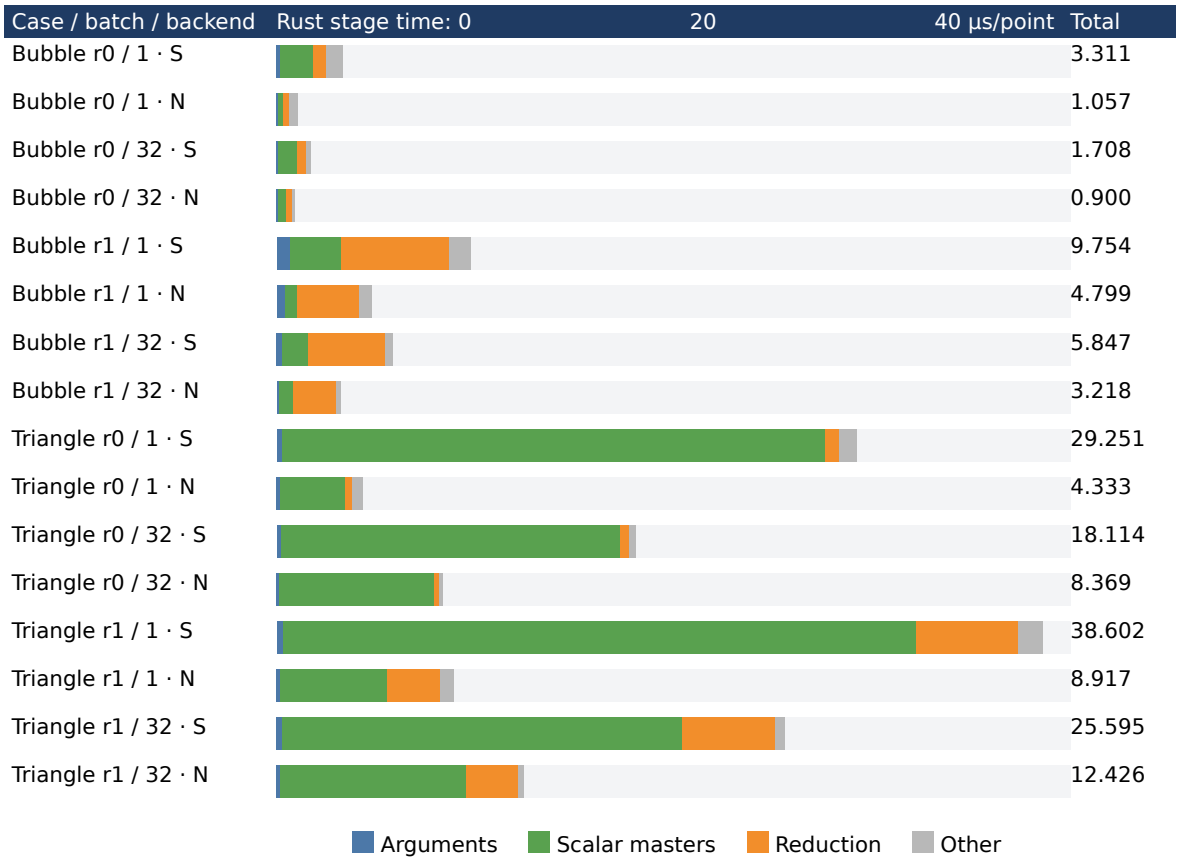

Both backends pass all eight numerical checks at $10^{-10}$. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; current-tree, higher-rank and physical performance remain unmeasured.

The newer optimized interpreter snapshot `5c9e3dd` passes eight numerical checks, with six timing regressions and two noisy, inconclusive rows. Its separate paired run is reported below. Subsequent default-stack, generalized-PV and vector-solve changes are unmeasured.

Newer validation includes four integrated graphs. The top pentagon at Float 128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. Physical timings are unmeasured.

## 2 Historical stage profiles: 4c32ece

Rust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference on page 1, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.




"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

### 2.1 Preparation and measurement conditions

| Backend | Bubble r0 | Bubble r1 | Triangle r0 | Triangle r1 |
|---|---|---|---|---|
| Interpreter | 1.572 s | 0.143 s | 0.104 s | 0.091 s |
| Native | 0.011 s | 0.012 s | 0.001 s | 0.013 s |

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. Rust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

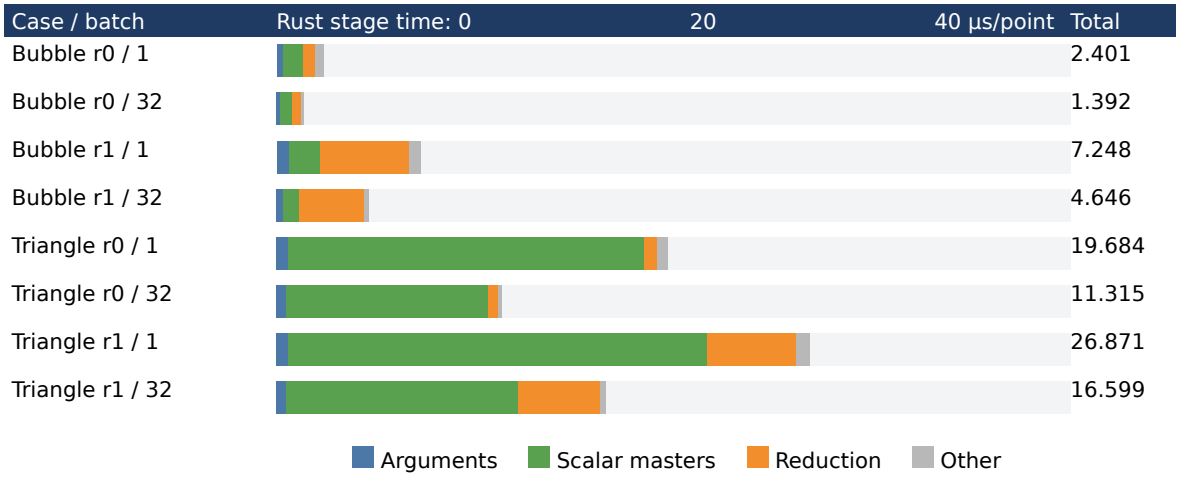
## 3 Newer kernel snapshot: `5c9e3dd`

All eight massive bubble/triangle checks pass the unchanged $10^{-10}$ criterion. Six timing rows regress; two exceed the allowed 2% relative confidence-interval width and remain **inconclusive**. Amber cells denote noise, not improvement. This commit predates default stack reuse, generalized PV and the vector-solve candidate. The overall performance requirement is unmet.

Table 2: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Each ratio uses its same-run paired original-IREGI measurement.

| Case | Rank | Batch 1 | Batch 32 |
|---|---|---|---|
| Bubble | 0 | 2.271 µs<br>IREGI 0.955 µs · ×2.381<br>99% CI 2.364–2.393<br>Regression | 1.389 µs<br>IREGI 0.901 µs · ×1.541<br>99% CI 1.535–1.546<br>Regression |
| Bubble | 1 | 7.146 µs<br>IREGI 3.019 µs · ×2.367<br>99% CI 2.362–2.384<br>Regression | 4.593 µs<br>IREGI 3.020 µs · ×1.521<br>99% CI 1.516–1.532<br>Regression |
| Triangle | 0 | 19.619 µs<br>IREGI 4.975 µs · ×3.948<br>99% CI 3.932–4.027<br>Inconclusive: noise | 11.310 µs<br>IREGI 4.948 µs · ×2.281<br>99% CI 2.263–2.296<br>Regression |
| Triangle | 1 | 26.724 µs<br>IREGI 10.231 µs · ×2.610<br>99% CI 2.597–2.624<br>Regression | 16.733 µs<br>IREGI 10.279 µs · ×1.620<br>99% CI 1.608–1.649<br>Inconclusive: noise |

### 3.1 Separately instrumented component timings




The stage sum belongs to a separate instrumented pass; “Other” is its total median minus named medians. These bars locate costs, not differences between revisions. Preparation is a single observation: bubble r0/r1 0.822/0.112 s; triangle r0/r1 0.067/0.061 s. It is excluded from the warm kernel ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs and all requested tensor slots. Structural allocations are reused; numerical results are recomputed. Transport is excluded. Rust 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`. All 391 source hashes, both binaries and the harness remain unchanged under the retained provenance checks. Absolute timings from earlier sessions cannot establish a revision speedup. Raw blocks, inputs, manifests and reproduction commands: `validation/benchmark/baseline-5c9e3dd/`.

# 4 Current implementation checkpoint

These experiments have separate source and binary identities. They extend the correctness evidence; they do not replace the historical timing tables.

## 4.1 Ordinary scalar-master storage

The default interpreter now reuses root temporaries through Symbolica's existing allocator. All five families preserve every output, precision and signed zero in 40 configurations each across f64, DoubleFloat and Float256. Shared helpers and Dual preparation are unchanged.

| Root temporary slots | A0 | B0 | dB0 | C0 | D0 |
|---|---|---|---|---|---|
| Before | 38 | 503 | 1,951 | 220 | 25,010 |
| After | 6 | 13 | 24 | 8 | 25 |

The default-stack workspace snapshot compiled with unchanged source hashes before the generalized-PV promotion below. Its current-guard acceptance passes 30 execution, 29 stability, six public mixed-precision and three scalar integration tests.

The paired DD E6 experiment preserves all three statuses and 12 complex Laurent outputs exactly, with the independent $10^{-23}$ check passing in both variants. Each uses order 128 and 3,720 D0 rows.

| Diagnostic observation | Before | After |
|---|---|---|
| Whole validation duration | 46.682 s | 43.822 s |
| Per-child peak resident memory | 2,052,724 KiB | 558,752 KiB |

**These are single-run diagnostics, with other compilation active; no timing gate or speed ratio is inferred.** The paired experiment uses an earlier, stricter Float guard; fixed-precision DD is unaffected. Current-guard tests are separate. Sources: `validation/stack-reuse/production-factory/`.

## 4.2 Generalized PV and unresolved precision

Generalized PV for $N \leq 4$ passes nine focused tests and 36 original-IREGI comparisons across 312 complex slots; maximum normalized error is $7.04 \times 10^{-12}$ at the unchanged $10^{-10}$ gate. The promoted PV and lazy-solve core also passes five supplemental behaviors: supplied products, multinomial weights, complex scalar/tensor comparisons and Hybrid isolation. The latter uses working128/256, requested53; its binary64 fixture failure is retained. These checks use frozen former-default master archives; combined default-stack integration is separate. Higher points retain IBP/E6. Sources: `validation/generalized-pv/` qualification, promotion and supplemental manifests.

The exact small-routing E6 oracle exposes a remaining production failure:

| Working bits | Requested bits | Public result |
|---|---|---|
| 128 | 128 | Unstable: nonfinite box output |
| 256 | 128 | Unstable: nonconvergent quadrature |

Both runs finish within their bounds and expose no successful values. No oracle comparison or requested-128-bit agreement is claimed. The frozen binary predates default stack reuse. See `validation/shifted-pentagon/production-series-qualification.json`.

# 5 Validation coverage

Table 3: **Evidence ledger.** "Source audited" records code inspection; "verified" identifies executed checks described in that row.

| Capability | Required evidence | Status |
|---|---|---|
| Independent reference oracle | Untouched/repaired Fortran twins<br>24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects. | Verified |
| Scalar-master component | OneLoopMaster A0/B0/C0/D0/DB0<br>Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit. | Verified |
| IBP component | Powers, dimension shifts and singular identities<br>21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots. | Verified |
| PV component | Every tensor moment and native Laurent slot<br>Canonical N1–4 ranks 0–7 pass. Generalized N≤4: nine focused + 36 original comparisons pass. Saved N5 rank-seven baseline: 1320 slots. Public N6/N7 tensor ranks pending. | Partial |
| Precision and Dual | binary64, DoubleFloat, Float, static/dynamic jets<br>Selected f64/DD/Float128–512 and static/dynamic jets pass. Mixed-precision guards: 29 + 6 pass. Small-routing E6 requested at 128 bits: Unstable. | Partial |
| Execution / CLI / core oracle | Batches through reduction and scalar masters<br>30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue. | Partial |
| Exceptional kinematics | Explicit errors and verified fallback<br>N5 rank-six rescue passes; saved rank-seven baseline passes 1320 slots. Legacy: 22 primary + 8 rescues. E6 failures and broader domains remain visible. | Partial |
| Graph/numerator preparation | FeynKit DOT to Symbolica coefficients<br>4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass. | Verified |
| Physical MadLoop comparison | Graph-isolated top loops and massless box<br>Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed. | Partial |

**Snapshot boundary.** Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Four physical triangle and five massive/complex box cases pass six-component mixed derivative checks. Historical timeouts remain visible. The N5 rank-six public rescue verifies all 210 moments / 840 slots at 128/256 bits, including an independent coefficient. Binary64 Rust and original IBP still exceed the fixed $10^{-10}$ criterion for that point. The frozen oracle passes 31 cases: 30 at binary64 and one generalized triangle through verified rescue. Broader domains remain pending.

## 5.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator $q_x^4$, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

| Method | Finite coefficient / $i$ |
|---|---|
| Parameter integral | 45.4316300015058951832997… |
| Original IBP | 45.43163000150602 |
| Original PV / hybrid | 46.11123944517427 |

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant Rust comparisons use original IBP and the independent parameter-integral result.

# 6 Measurement protocol

Pin release builds to one CPU core and record thread counts, hardware, OS, toolchain, flags, dependency/input hashes, seeds and warm-up. Identify cold or warm caches. Retain raw repeated-block observations, median, dispersion and an output checksum.

Preparation includes graph import, numerator construction and evaluator compilation. Time numerator, reduction, masters and total independently; summed instrumented stages are only diagnostic. Separate library costs from subprocess/file I/O and match physical routing, modes and stage boundaries.

Use identical samples at batch widths 1, 4, 8, 16, 32 and 64, matching numerical precision and including rescue costs. Retain failed lanes. Every speed gate requires correctness: stable binary64 uses the fixed $10^{-10}$ mixed tolerance on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput per topology/rank/mass family, with IBP, PV and hybrid separate. The target envelope includes seven propagators, rank seven, generalized powers and dimension shifts.

The kernel harness now records every requested configuration before starting subprocesses, preserving interrupted and unrun rows. The retained Python suite passes 37 ledger, provenance, protocol, transport and comparison boundary checks, without numerical backends. Explicit matched manifests identify the separately executed `5c9e3dd` timing snapshot.

An identified debug binary passes the $gg \to H$ harness smoke at batches 1 and 3: all 15 moments / four slots agree with three IREGI modes within $3.55 \times 10^{-15}$, and MadLoop within $8.02 \times 10^{-15}$. Seven-stage protocol checks pass; short debug timings, live-source provenance and repeated-point scaling remain unqualified. See `validation/benchmark/physical-smoke/qualification.json`.

Actual concurrent evaluation remains unqualified under the installed one-thread Symbolica license; the guarded two-worker test is ignored. The corrected sequential corpus passes, using scalar D0 with a tensor triangle and second-order Dual bubble. Its original rank-one f64 box rejection remains recorded; sequential agreement does not establish concurrency. See `validation/concurrent-workspaces/qualification.json`.

# 7 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `typst compile docs/benchmark.typ docs/benchmark.pdf`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

## 7.1 Validation reproduction and remaining checks

Run recorded commands with the pinned environment from `scripts/env.sh`. Exact commands and source hashes accompany each JSON result. The frozen 31-case oracle includes one explicit 128/256-bit rescue; all 30 legacy fixtures pass through 22 primary results and eight verified rescues. Three active rescue, four bubble-Taylor, three scalar API and nineteen CLI smoke checks pass. The new 68-test default-stack acceptance is a separate source scope; earlier failures remain retained.

The final frozen master source passes 27 checks, including extreme-scale B0, six-component scale-only jets in all families, mixed input precision and the full physical D0 derivative batch; see `validation/scalar-masters/qualification.json`. The separate exact factory-parity/default-stack qualification is described above. Generated native kernels remain opt-in; no new timing gate passes.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles disabled/enabled: 480 evaluations and 1920 slots. Nine extracted FeynKit partial-fraction and two planar formula tests pass; public integration remains pending.

Exact PV coefficients with Fortran leaves pass N1–4 through rank seven. For N6, inline native-master 128/256-bit arrays converge over 1320 complex slots within $6.46 \times 10^{-29}$; retained binary64/original results fail. Full public N6/N7 tensor arrays remain unqualified. The latest N6 rank-seven probe times out during workspace preparation at 180 s, after a 107.836 s plan with 330 moments, 31,670 aliases and 56 masters; no values are produced. See `validation/public-highpoint/`. The saved N5 rank-seven baseline passes separately; historical evidence remains in `data/results.json`.

# 8 Physical fixture qualification

| Single graph | Pointwise numerator | Integrated Rust result |
|---|---|---|
| Top triangle: $gg \to H$ | Verified | Verified |
| Top box: $gg \to HH$ | Verified | Verified |
| Top pentagon: $gg \to HHH$ | Verified | PV/hybrid + MadLoop |
| Massless up-quark box | Verified | Verified |

## 8.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five schema-2 captures under `validation/physical/madloop/fixtures/` contain 40 graph records, including a second triangle helicity. All thirteen preparation tests and four selected physical numerator comparisons pass; this is not 40 integrated validations. Actual evaluated momenta are exported because MadLoop can correct the requested phase-space point. The compact triangle reference is `validation/physical/ggh-numerator-reference.json`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within $1.86 \times 10^{-15}$ normalized error and MadLoop within $9.38 \times 10^{-15}$, including poles. Its fixed-color raw finite value is $-147.86843349622455$, normalized to $-0.9363877940734862$. Values and executable provenance are retained in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. All nineteen current CLI smoke checks pass. Reference data and comparison arithmetic are binary64; higher working precision certifies no extra digits. Short unpinned timing checks are excluded from benchmark evidence. The top-box binary64 rejection remains recorded: PV fails its cancellation guard and IBP fallback is nonfinite. Complete Float128 and Float256 CLI reruns now pass all 280 tensor slots against all three reference modes, within $1.15 \times 10^{-13}$; MadLoop contraction error is at most $4.77 \times 10^{-15}$. Each run freshly prepares/evaluates the numerator. Their normalized tensor difference is $6.09 \times 10^{-37}$, with identical contracted values after binary64 conversion. That comparison arithmetic certifies no extra digits. Evidence is retained in `validation/physical/gghh-cli-precision-convergence.json` and its linked Float 128/256 reports. A model-import regression was resolved by a narrow Symbolica-backed quantum- number adapter. Four boundary, thirteen preparation and four physical numerator tests pass with the fix; other numerical tokens retain their precision.

The current optimized CLI also passes the massless up-quark box. All 70 moments / 280 slots agree with each original mode: maximum error $2.85 \times 10^{-14}$ against IBP and $7.11 \times 10^{-15}$ against PV/hybrid. The isolated MadLoop finite/single/double coefficients agree within $6.74 \times 10^{-15}$, with fixed-color factor $-\frac{1}{8}$. Evidence is retained in `validation/physical/gggg-up-cli-comparison.json` its one-run wall time is not a qualified performance measurement.

Complete top-pentagon CLI reruns at Float 128/256 now agree with original PV/hybrid for all 280 slots within $4.63 \times 10^{-14}$, and with MadLoop within $9.30 \times 10^{-14}$. Between-run tensor difference is $7.79 \times 10^{-17}$ under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

## 8.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error $5.47 \times 10^{-28}$. One equal-complex D0 case also passes, with error $1.11 \times 10^{-34}$. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error $2.53 \times 10^{-30}$ under the unchanged $2^{-70}$ criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

# 9 Open qualification ledger

| Unresolved item | Observed evidence and remaining check |
|---|---|
| Binary64 baseline accuracy | Historical Rust error $2.23 \times 10^{-10}$ and original IBP error $3.40 \times 10^{-10}$ fail the unchanged gate. Public 128/256-bit rescue now passes at the retained rank-six point. |
| Broader pentagon rescue | Rank six passes 840 slots; the saved vendored baseline passes 1320 rank-seven slots and one independent coefficient. Later precision/routing edits and broader domains remain unqualified. |
| Physical pentagon reference | Complete Float 128/256 graph evaluations pass PV/hybrid and MadLoop. Original IBP still disagrees; all-mode reports remain failed. Binary64 and broader kinematics remain unqualified. |
| Shifted scalar domain | Thirteen standalone checks and public finite fixtures pass. The paired DD stack candidate also passes exactly. Small-routing E6 at working128/256 and requested128 returns Unstable; broader precision/derivative domains and native E6 remain unqualified. |
| Broader derivative domains | The full five-case mixed D0 batch now passes with error below $2^{-70}$. True thresholds and unsupported transverse massless/Gram/Cayley domains retain typed errors; old timeouts remain. |
| Physical top-box binary64 | The original binary64 guard rejection remains visible. Complete Float 128/256 evaluations now pass; this does not establish binary64 stability or performance. |
| Supplied-product high points | Certified N6 scalar: working128/256, requested53, zero poles and invalid-lane isolation pass. Both binary64 variants reject at the same guard. Independent quadrature differs by $1.53 \times 10^{-23}$, below estimated uncertainty $6.11 \times 10^{-22}$ (not rigorous). N7/tensor/mixed-Dual coverage remains pending. |
| Legacy snapping scope | All 30 retained cases pass: 22 primary plus eight verified 128/256-bit rescues. Earlier failures are preserved; generalized requests and broader domains remain unqualified. |
| Mixed-precision accuracy | Three retained false-success cases are repaired: 29 stability and six public tests pass. The active-master guard remains conservative, its estimates noncertified and its overhead unmeasured. |
| Runtime performance | Interpreter: 7 regressions/1 improvement. Native: 5 regressions/1 parity/2 improvements, in a separate paired session. Newer5c9e3dd: 6 regressions/2 noisy inconclusive rows. Subsequent changes remain unmeasured; diagnostic memory observations do not satisfy the gate. |

## 9.1 Canonical physical reference audit

Original IBP disagrees with PV/hybrid in $gg \to HHH$ despite stable flags; maximum normalized finite error is 0.600134. Other three families agree within $2.22 \times 10^{-14}$; see `validation/physical/canonical-reference-audit.json`. The next page independently adjudicates moment 54. Reference stability does not certify Rust.

## 9.2 Independent integral evidence

Independent 50/80-digit rank-four bubble integrals and 60/100-digit lightlike bubble integrals with derivatives are retained under `validation/pv-high-rank`. Correctly rounded masters reduce synthetic pentagon error to $3.11 \times 10^{-12}$ with Float256 reduction or $2.23 \times 10^{-11}$ with binary64 reduction, isolating amplified master error.

The equal-mass D6 series oracle has a proven remainder below $2^{-321}$; its production failures remain visible above. The new N6 scalar oracle and lazy Cayley solve evidence are retained under `validation/invariant-certificate/solve-vector/`. Neither establishes broader high-point tensor or performance coverage.

# 10 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

## 10.1 Rank-six Euclidean tensor

For squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct simplex integration gives native finite coefficient $-0.0024283789354471 23i$. Independent coordinate mappings agree within $4.34 \times 10^{-19}$; reported quadrature error is at most $2.77 \times 10^{-15}$. The bound $F \geq 3.75$, angular integration and power counting exclude poles. This uses no reduction/master library; numerical error estimates are not interval certificates. See `validation/pv-high-rank/pentagon-simplex-oracle.json`. Historical binary64 errors remain $2.23 \times 10^{-10}$ for Rust and $3.40 \times 10^{-10}$ for IREGI. Public rescue flags the primary unstable and verifies all 840 slots at 128/256 bits plus this coefficient below $10^{-13}$. `validation/rescue-pentagon-rank6.json` retains the binary and assertions; full arrays were not printed and its exact build-source snapshot was not saved.

The coherent vendored Symbolica baseline passes four internal scope tests, 57 focused core tests and all 1320 rank-seven slots at 128/256 bits, including the independent coefficient. Its saved source predates later precision/routing edits. Current workspace compilation passes with all 329 source/manifest hashes unchanged; later runtime checks remain separate. The 109.44 s cold run is diagnostic, not a performance benchmark. See `validation/symbolica-production-qualification.json`. The earlier 240 s timeout and 115.35 s isolated-patch run remain historical.

## 10.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree $-2$; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from $7.9 \times 10^{-31}$ to $2.3 \times 10^{-69}$, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

## 10.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

| Method | Finite $J_1$ |
|---|---|
| Independent simplex | $-0.00017714013826400 81806076521310$ |
| Original scalar/general APIs | $-0.00014953025024128455$; every mode reports stable |

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within $4.93 \times 10^{-15}$. A box-only half-line implementation also agrees to approximately $8.42 \times 10^{-17}$ or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. Thirteen consolidated D6 tests cover eight genuine-master checks and five endpoints. Retained public f64 fixtures pass finite values, poles, scaling, rejected endpoints and lane isolation; the earlier quadrature-budget failure remains historical. Pre-vendor public DoubleFloat and Float128 tests now pass three scaled lanes at requested 80-bit accuracy and $10^{-23}$ normalized oracle tolerance. First common-mass Dual primal/derivative checks pass at $2 \times 10^{-14}$, with zero poles. The DD/Float validation durations total 165.69 s, exceeding the historical 150 s combined timeout; this is no performance benchmark. Build identities and limits are in `validation/shifted-pentagon/public-core-qualification.json`. A separate vendored paired DD E6 test passes exact stack-allocation parity. Full requested 128-bit accuracy fails for the new small-routing fixture; higher/mixed Dual, broader endpoints and native E6 remain unqualified. The original complex-box reference discrepancy is documented in `validation/shifted-pentagon/contours.json`.

# 11 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

## 11.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order $(p^0,p^1,p^2,p^3)$. Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of $\pi$. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes $\mu^2$ to QCDLoop and $\mu$ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

## 11.2 Native poles and MadLoop conversion

The native result order is

$$(U, I, J, F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank $r$ to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master $(L_0, L_1, L_2)$, the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U, I, J, F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

## 11.3 Polynomial and tensor interfaces

A rank bound $R$ has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of $q^0$, $q^1$, $q^2$ in descending order; the exponent of $q^3$ is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

# 12 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

## 12.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

## 12.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep $D$ symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

## 12.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

**Primary references:** Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.